Assignment 1 - Phase 3 - Saurav Dasgupta C52

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1. Find the Answer

Problem Statement (P0)

Target Segment

Knowledge seekers such as students, professionals and lifelong learners who come to YouTube with a specific question and expect to find a clear answer.

P0

Knowledge seekers often come to YouTube with a specific question in mind. While YouTube is excellent at helping users discover videos, it does not help them discover answers.

Today, users face two key challenges:

1. When they search for a question, YouTube returns a list of videos instead of directly helping them identify which videos actually contain the answer they are looking for.
2. Even after opening a relevant video, users often spend several minutes scrubbing through long lectures, podcasts or interviews before reaching the part that answers their question.

For users whose primary intent is to learn something quickly, this creates unnecessary effort and often leads to abandoning videos before finding the answer.

Why does this problem happen?

Based on the research conducted in Phase 2, I identified four underlying reasons.

1. Search retrieves videos instead of answer moments

The ranking system is designed to recommend complete videos rather than the specific moments inside those videos that answer a user's question.

2. Long-form educational content is difficult to navigate

Educational videos, lectures and podcasts are often more than an hour long. Even when timestamps exist, users still spend considerable time locating the exact section that answers their question.

3. Users cannot judge relevance before clicking

The search results provide information such as title, thumbnail and duration, but they do not explain whether the video actually answers the user's query.

4. Informational intent and entertainment intent share the same search experience

Someone searching for *"How fusion reactors work"* has a very different goal from someone searching for a movie trailer or music video. However, YouTube currently treats both searches in exactly the same way.

Solution Exploration

I explored multiple ideas before selecting the final MVP solution.

Idea	Root Cause Addressed	Key Assumption	Confidence
AI-generated answer above search results	Users need direct answers	Users trust AI summaries over creator videos	Low
Jump-to-answer timestamps	Navigation inside videos	Relevant timestamps can be identified accurately	High
Question-aware video chapters	Long-form navigation	Chapters align with user questions	Medium
Highlight answer sections inside the player	Navigation after playback begins	Users first choose the correct video	High
Search by transcript	Better retrieval	Transcript quality is consistently reliable	Medium
Community-curated answer timestamps	Better answer discovery	Users are willing to contribute consistently	Low
Answer Mode (Selected)	Solves all four root causes	Relevant answer sections can be identified with sufficient accuracy	High
Cross-video AI answer summary	Faster learning	AI can summarise creator content fairly	Moonshot

Idea	Root Cause Addressed	Key Assumption	Confidence
Topic-based answer filters	Better answer refinement	Topics can be extracted reliably	Medium
Personalized answer ranking	Better recommendations	User behaviour improves future rankings	Medium

Prioritizing the Solutions

To compare the shortlisted ideas objectively, I used the **RICE framework**.

Factor	Meaning
Reach	How many users within the target segment would benefit from the solution.
Impact	How significantly the solution improves the user experience if it succeeds.
Confidence	How confident I am that the underlying assumptions are correct based on available research.
Effort	Relative implementation effort. Lower effort leads to a higher RICE score.

For this exercise, each dimension was scored on a scale of **1–10**.

The RICE score is calculated as:

(Reach × Impact × Confidence) ÷ Effort

The score helps compare ideas objectively, but the final decision also considers whether the solution directly addresses the identified P0.

Shortlisted Solutions

Solution	Reach	Impact	Confidence	Effort	RICE Score
Answer Mode	9	10	8	6	**120**
Jump-to-answer timestamps	8	8	7	6	74.7
Topic-based answer filters	6	6	7	5	50.4
AI-generated answer summary	7	9	5	8	39.4

Answer Mode received the highest score because it addresses every major pain point identified during Phase 2 while remaining feasible as an MVP.

Why I selected Answer Mode

Answer Mode is an optional search experience designed specifically for users with informational intent.

When enabled, YouTube no longer returns only complete videos. Instead, it identifies the most relevant sections across long-form videos that answer the user's question.

Rather than replacing videos with AI-generated answers, it helps users discover the most relevant parts of creator content more efficiently while still providing a clear path to the complete video.

The Answer Mode Experience

Each search result is represented as an **Answer Card** instead of a traditional video card.

Each card contains:

- The most relevant answer section from the video
- A short AI-generated summary
- A relevance label (**Best Match**, **Highly Relevant**, or **Relevant**)
- A **Why this section matched** explanation describing why the section was selected
- **Watch This Moment**, which starts playback directly from the recommended timestamp
- **Watch Full Video** for users who wish to continue watching the entire content
- **More Relevant Sections**, allowing users to explore additional matching moments from the same video

When users expand **More Relevant Sections**, each additional section displays its own summary, relevance label and explanation.

Helping Users Discover Answer Mode

Answer Mode is intentionally designed as an optional experience rather than replacing YouTube's existing search behavior.

If a user searches with Answer Mode turned off, YouTube continues returning standard video results.

However, when the search query appears informational, the platform displays a contextual recommendation card introducing Answer Mode.

> **Looking for a direct answer?**
>
> Answer Mode helps you:
>
> ✓ Jump directly to the most relevant moments
>
> ✓ Understand why each section matches your question
>
> ✓ Continue watching the full video whenever you want
>
> **Turn on Answer Mode**
>
> **Not now**

Selecting **Turn on Answer Mode** immediately refreshes the search results using the new experience.

Selecting **Not now** dismisses the recommendation and continues showing the standard search experience.

This keeps the existing YouTube experience unchanged while making the feature discoverable only when it is likely to help.

Enhanced Video Player

After selecting **Watch This Moment**, playback begins directly from the recommended section.

The Video Player has been enhanced to make answer navigation easier.

It includes:

- Highlighted sections on the timeline showing every relevant answer segment
- A **Relevant Moments** panel listing all matching sections
- AI-generated summaries for each section
- Relevance labels (**Best Match**, **Highly Relevant**, **Relevant**)
- **Why this section matched** explanations
- Thumbs Up / Thumbs Down feedback to improve future rankings
- A **Watch Full Video** option for uninterrupted viewing

This allows users to move between relevant sections without repeatedly scrubbing through the video.

Why the Other Shortlisted Solutions Were Not Selected

Jump-to-answer timestamps

This idea improves navigation only after a user has already opened the correct video.

It does not solve the earlier problem of helping users identify which videos actually contain the answer they are looking for. As a result, it addresses only part of the journey.

Topic-based answer filters

Topic filters would allow users to refine answer sections based on concepts such as *Tokamak Reactors* or *Plasma Confinement*.

While useful, they depend entirely on Answer Mode first identifying relevant sections. They improve exploration rather than solving the core discovery problem, making them better suited for a later

release.

AI-generated answer summary

A single AI-generated answer across multiple videos could reduce the amount of content users need to watch.

However, this approach introduces several concerns:

- It shifts value away from creator content.
- It increases the risk of hallucinated or incomplete answers.
- It fundamentally changes YouTube from a video discovery platform into an answer engine.

For the MVP, I wanted the solution to improve discovery rather than replace videos altogether.

Why the Remaining Ideas Were Rejected

Idea	Reason for Rejection
Search by transcript	Improves retrieval quality but does not significantly improve the overall user experience.
Question-aware chapters	Depends heavily on creators organising content consistently.
Community-curated timestamps	Difficult to scale and relies on continuous community participation.
Personalised answer ranking	Requires substantial behavioural data before providing meaningful value.
Cross-video AI summary	Considered a long-term vision due to technical complexity and trust concerns.

Why Answer Mode Stands Out

Among all the ideas explored, Answer Mode was the only solution that addressed the complete user journey.

It helps users:

- Identify which videos contain answers to their question.
- Understand why a section has been recommended.
- Jump directly to the most relevant moment.
- Continue watching the complete video if they choose.
- Improve future recommendations through simple feedback.

Most importantly, it enhances YouTube's existing value proposition instead of replacing it. Users continue discovering creator content, but they reach the answers they need much faster.

Ideas Deferred Beyond the MVP

To keep the MVP focused, several ideas have been intentionally deferred to future iterations.

These include:

- **Next Best Answer**, allowing users to jump directly to the best matching section in the next recommended video.
- **Refine Your Answer**, enabling users to filter answer sections using topic labels extracted from the matched content.
- **Cross-video AI summaries**, synthesizing insights across multiple creators.
- **Answer Section History**, allowing users to revisit previously watched answer moments.
- **Personalized answer ranking**, using historical learning behavior to improve future recommendations.

These ideas build naturally on the core Answer Mode experience but depend on first validating that users find value in section-based answer retrieval.

2. Prototype, Validation Plan & Outcome Measurement

Prototype

Prototype Overview

To validate the proposed solution, I created a clickable prototype demonstrating the complete end-to-end Answer Mode experience. Instead of prototyping the entire YouTube application, I focused only on the user journey required to validate the core hypothesis:

> **Can YouTube help users find answers inside videos instead of making them search through videos?**

The prototype covers the journey from searching for a question to consuming the answer inside the video player.

Prototype Flow

Screen 1 – YouTube Home

The journey begins on the existing YouTube homepage.

Users can search exactly as they do today.

A new **Answer Mode** toggle has been introduced beside the search bar.

By default, Answer Mode remains **OFF**, preserving YouTube's existing search behaviour.

Screen 2 – Discovering Answer Mode

If a user searches for an informational query while Answer Mode is turned OFF, YouTube initially returns the standard video results.

A recommendation card appears at the top of the search results explaining Answer Mode.

The card briefly explains that Answer Mode can:

- Jump directly to relevant moments
- Explain why each section matches
- Allow users to continue watching the full video

The user can either:

- **Turn on Answer Mode**
- **Not now**

Selecting **Not now** dismisses the card and continues with the existing search experience.

Selecting **Turn on Answer Mode** refreshes the search results using Answer Mode.

This ensures users discover the feature naturally without changing YouTube's default behaviour.

Screen 3 – Answer Mode Search Results

Instead of returning only videos, the search results now surface the most relevant answer sections across videos.

Each result is displayed as an **Answer Card**.

Every Answer Card contains:

- Video title
- Creator information
- Thumbnail
- AI-generated summary of the recommended section
- Relevance label:
  - **Best Match**
  - **Highly Relevant**
  - **Relevant**
- **Why this section matched**
- **Watch This Moment**
- **Watch Full Video**

If multiple relevant sections exist within the same video, only the strongest section is shown initially.

A button labelled

"View X more relevant sections"

allows users to expand additional matching sections from the same video.

Screen 4 – Expanded Relevant Sections

When the user expands the additional sections, every matching interval becomes visible.

Each interval includes:

- Timestamp
- Short AI-generated summary
- Relevance label
- Why this section matched
- Watch This Moment

This allows users to compare multiple answer sections before deciding which one to watch.

Screen 5 – Enhanced Video Player

Selecting **Watch This Moment** opens the video directly at the beginning of the chosen answer section.

The Video Player has been enhanced with several new capabilities.

Timeline Highlights

Relevant answer sections are highlighted directly on the playback timeline, allowing users to identify every answer-bearing segment at a glance.

Relevant Moments Panel

A side panel lists every relevant section within the current video.

Each section displays:

- Timestamp
- AI-generated summary
- Relevance label
- Why this section matched

Selecting any section immediately jumps playback to that point.

Feedback

Each answer section contains

👍 Helpful

👎 Not Helpful

This feedback helps improve future ranking quality.

Continue Watching

If users become interested in the broader context, they can continue watching the entire video without interruption using **Watch Full Video**.

This preserves YouTube's existing creator-first experience.

Paper Sketch / Flow Diagram

The prototype was initially designed as a simple user flow before building the interactive prototype.

Here's the PowerPoint presentation detailing the flow diagram (attached as a pdf) :

attachment:3fc7b5e0-0a61-4e45-b75f-5011e32e5816:Answer_Mode_Case_Study.pdf

Prototype Link

Live Prototype :

https://claude.ai/public/artifacts/72535428-7de1-4141-bb98-f633c92ec44d

Walkthrough Video

A 2–5 minute walkthrough accompanies the prototype.

The walkthrough demonstrates:

- the problem being solved,
- Answer Mode discovery,
- Answer Mode search,
- expanding relevant sections,
- jumping directly to answer moments,
- enhanced navigation inside the Video Player.

Walkthrough Video

> https://drive.google.com/file/d/1Q0cpYMpVCsXNFHsIw1qjgwx_w0tO8e2g/view?usp=drive_link

3. Plan the Outcome

The objective of the MVP is to validate whether Answer Mode helps knowledge seekers reach relevant answers faster than the existing YouTube search experience.

Success Metrics

Metric Type	Metric	Why?
Primary Metric	Percentage of informational searches where users successfully watch a recommended answer section	Directly measures whether users are finding and consuming the answers surfaced by Answer Mode.
Secondary Metric 1	Percentage of users who choose **Watch This Moment**	Indicates whether the surfaced answer sections are compelling enough to replace the existing behavior of opening entire videos.
Secondary Metric 2	Percentage of users expanding **View More Relevant Sections**	Measures whether users find value in exploring multiple answer sections instead of accepting only the top recommendation.
Guardrail Metric	Percentage of users continuing to watch the full video after entering through Answer Mode	Ensures Answer Mode improves answer discovery without reducing creator engagement. Maintaining healthy full-video watch behaviour confirms the feature complements YouTube's ecosystem rather than replacing it.

Risk Assessment

Assumption	Failure looks like	Likelihood	Impact	Early Signal	Mitigation
Answer extraction identifies relevant sections accurately	Users frequently skip recommended sections or abandon playback	Medium	High	Low Helpful feedback and low Watch This Moment completion	Launch with educational content where transcript quality is highest and improve ranking iteratively.
Users understand Answer Mode	Low adoption despite informational searches	Medium	Medium	Few users enable Answer Mode	Introduce contextual recommendation cards for informational searches and

Assumption	Failure looks like	Likelihood	Impact	Early Signal	Mitigation
					iterate based on user research.
Relevance labels build trust	Users ignore Best Match recommendations	Low	Medium	Similar click rates across all labels	Simplify wording and improve explanation quality through usability testing.
User feedback improves ranking quality	Helpful/Not Helpful feedback is rarely used or inconsistent	Medium	Medium	Low feedback submission rate	Use implicit behavioral signals alongside explicit feedback to improve recommendations.

MVP Scope

Included

- Answer Mode toggle
- Contextual onboarding card
- Answer Cards
- Best Match / Highly Relevant / Relevant labels
- Why this section matched
- Watch This Moment
- Watch Full Video
- Multiple relevant sections
- Enhanced Video Player
- Timeline highlights
- Relevant Moments panel
- Section-level feedback

Out of Scope

The following ideas are intentionally excluded from the MVP:

- Next Best Answer across videos
- Refine Your Answer filters
- Cross-video AI summaries

- Personalized answer ranking
- Answer History
- Learning collections

These ideas will only be explored after validating that users find value in section-based answer retrieval.

Validation Hypothesis

The MVP is based on the following hypothesis:

> **If YouTube surfaces the most relevant answer sections instead of only complete videos, knowledge seekers will reach useful answers faster while continuing to engage with creator content.**

The success metrics and prototype have been designed specifically to validate this hypothesis before investing in more advanced discovery and personalization features.

4. Product Requirement Document

Assignment_1_-_Product_Requirement_Document_-_YouTube_Answer_Mode_-_Saurav_Dasgupta_C52.pdf